

Networking Overview

What is Networking?

Networking, in the context of computer science, refers to the process of connecting multiple computers or devices to share resources, exchange data, or communicate with each other. It enables devices to work together, whether they are in the same room or across the globe, through wired or wireless connections.

Theoretical Concepts

Definition: Networking links devices to exchange data using hardware (routers, cables) and software (protocols like TCP/IP).

Purpose: Share files, enable communication (e.g., email), and transfer data securely.

Key Components:

Devices: Computers, routers, switches.

Connections: Cables or Wi-Fi.

Protocols: Rules like TCP/IP, HTTP.

Network Types: LAN (local, e.g., home), WAN (wide, e.g., internet), MAN, PAN.

Models:

Client-Server: Server provides services to clients (e.g., web hosting).

Peer-to-Peer: Devices share directly (e.g., file sharing).

Hardware Requirement for Networking

1. End Device (System/Machine/Laptop/Tablet)
2. Network Interface Card (NIC)
3. Cables & Connectors
4. Hub
5. Switch
6. Router
7. Access Point (AP)

End Devices

What are End Devices?

- End devices, also called hosts, are the computers, gadgets, or machines that use the network to communicate.
- They are called “end devices” because they sit at the edge of the network — they are not responsible for forwarding data like routers or switches.
- **Purpose:**
 - End devices send, receive, and process data over the network.
 - They are the source or destination of network communication.
- **Examples of End Devices:**
 - Computers / Laptops
 - Smartphones / Tablets
 - Servers: Printers / IP Cameras / IoT Devices



Network Interface Card (NIC)

What is it?

A NIC is a firmware card (or built-in chip) in your computer that connects it to a network, like plugging your device into the internet or a local network using Ethernet or Wi-Fi.

- Each NIC has a unique MAC address.
- Can be wired (Ethernet) or wireless (Wi-Fi adapter)

Interesting Use (Why it's cool)

Without a NIC, your computer couldn't browse the internet, stream videos, or share files with other devices! It's like the computer's "mouth" for network communication.

Real-World Example

Your laptop's Wi-Fi card (a NIC) connects to your home Wi-Fi to let you watch YouTube. In RHEL, check it with: `ip link show`



NIC Card Types:

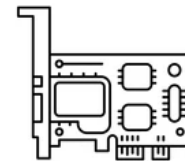
- Wired NIC -----→ RJ45, Internal NIC
- Viewless NIC -----→ WIFI NIC, Adapter
- Virtual NIC -----→ Used in a Virtual Machine

Types of IP allocation to a NIC

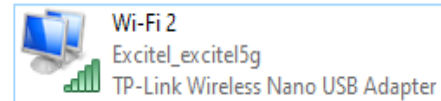
- By DHCP (Automatic received by the Router)
- By Static (You choose an IP from an IP Pool range)

A NIC has the following

- IP (Public IP & Private IP as per connection)
- MAC Address (Physical address)

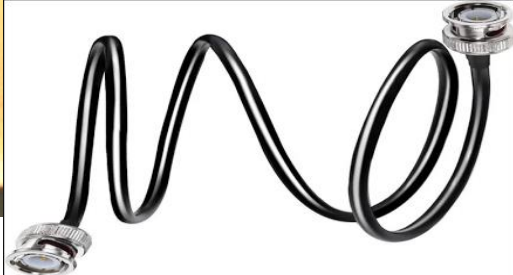
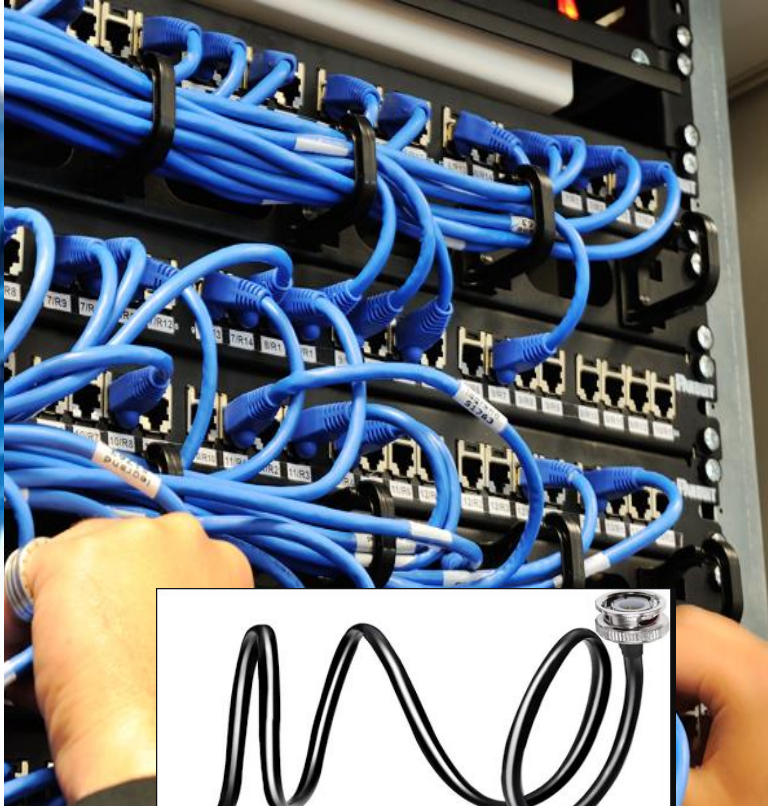


192.168.1.1



I
n
t
e
r
n
e
t
.

• Coaxial Cables:



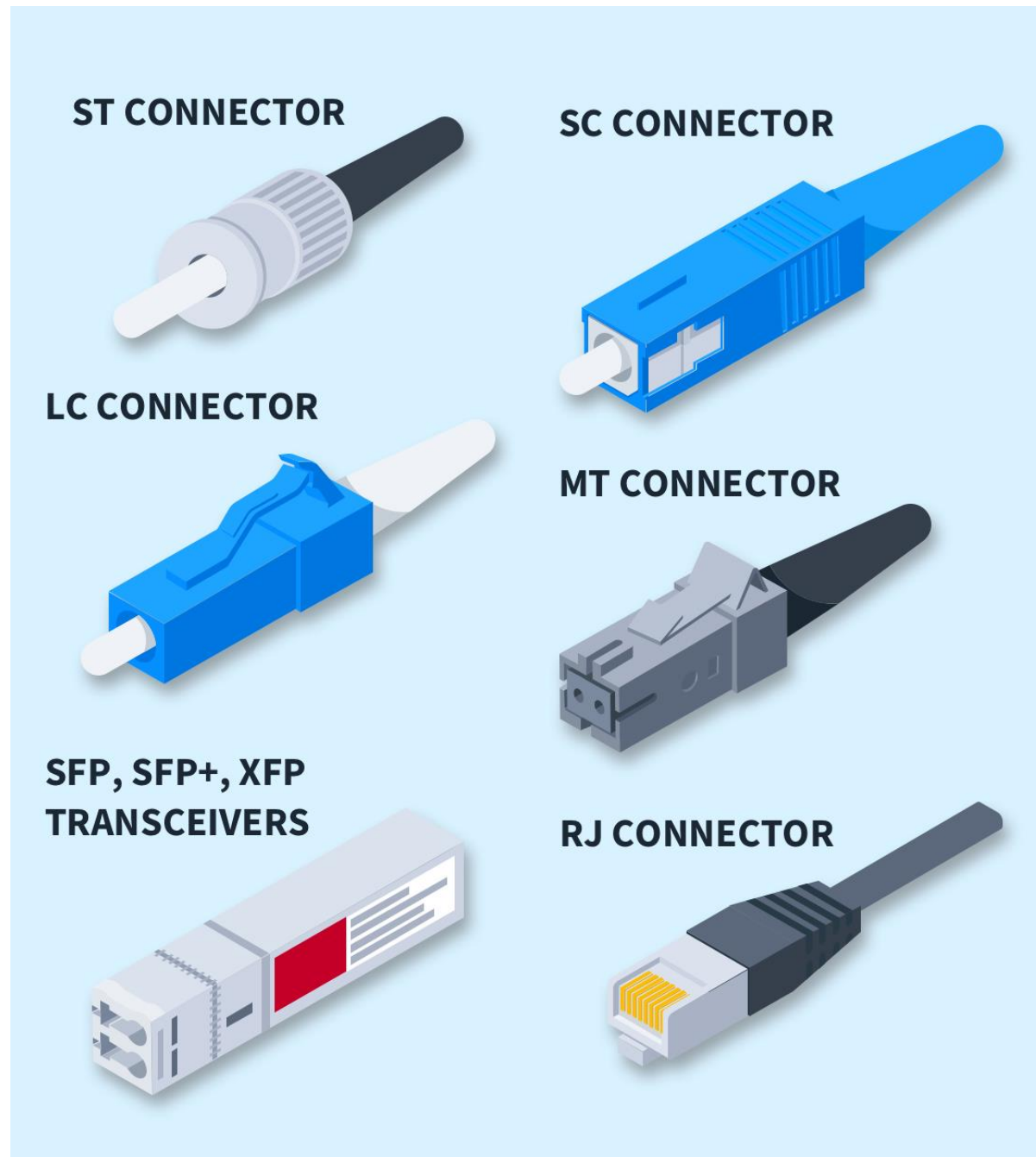
Network Connectors?

Connectors are the plugs or ports at the end of network cables that physically link cables to devices like NICs, routers, switches, hubs, or access points. They ensure data signals are transmitted between devices.

Types of Connectors

Each cable type uses a specific connector designed for compatibility.

- **RJ45** for Ethernet cable connectors
- **LC/SC** for Fibre optic cable connectors
- **Coaxial (BNC, F-Type)** for Coaxial cable connectors



Hub

What is it?

A hub is a simple device that connects multiple devices in a network, but sends data to all devices, like shouting in a room. This was a dumb device that supported no privacy.

Interesting Use (Why it's cool)

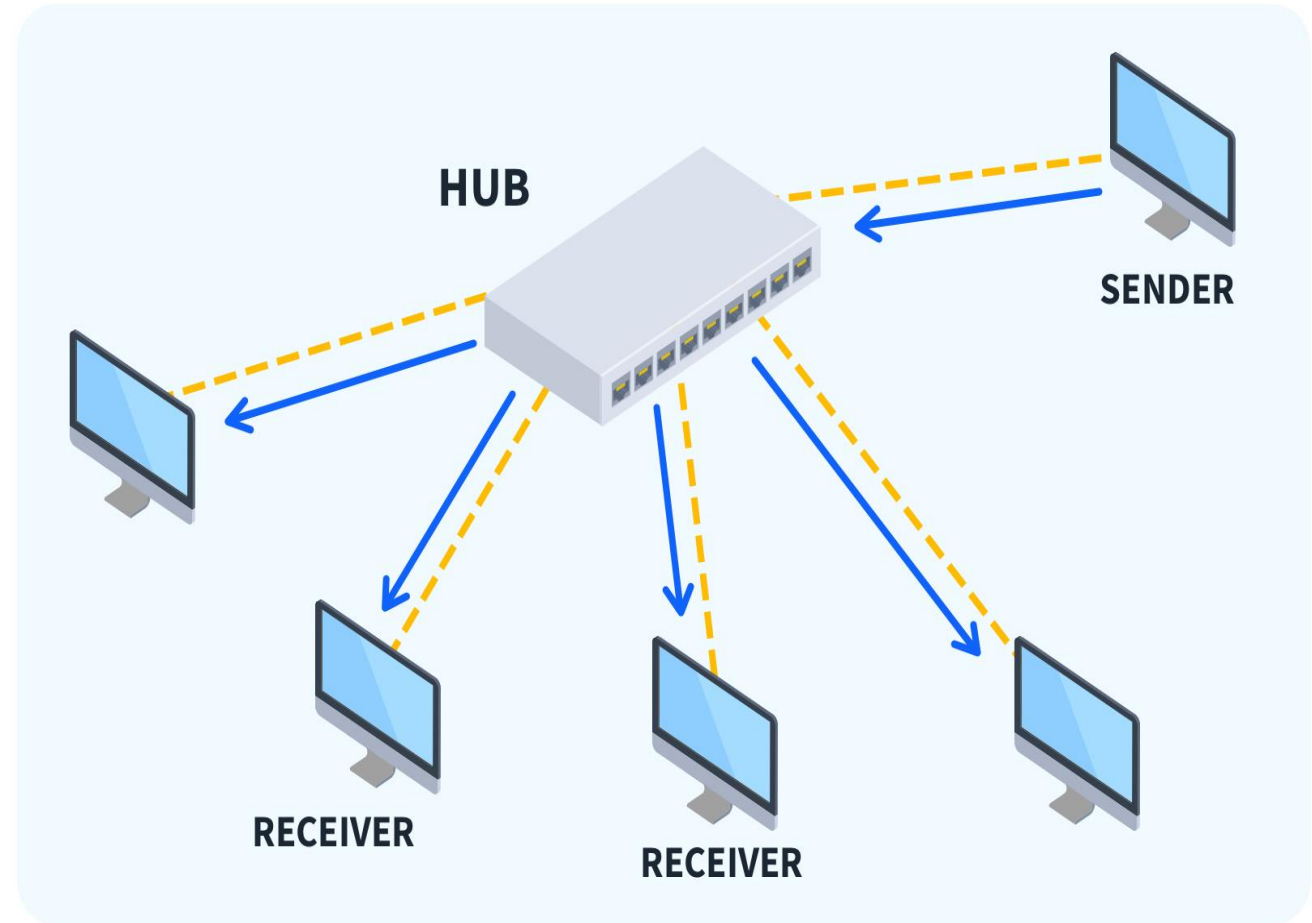
Hubs are old-school and less common now, but they're super simple to set up—just plug and play, no configuration needed!

Works on Which Layer?

OSI Layer 1 (Physical: just passes signals without processing).

Real-World Example

A hub might connect old printers in a small office, sending data to all.



Switch

What is it?

A switch connects devices in a single network (like in an office) and sends data only to the right device, like a smart mail sorter.

Interesting Use (Why it's cool)

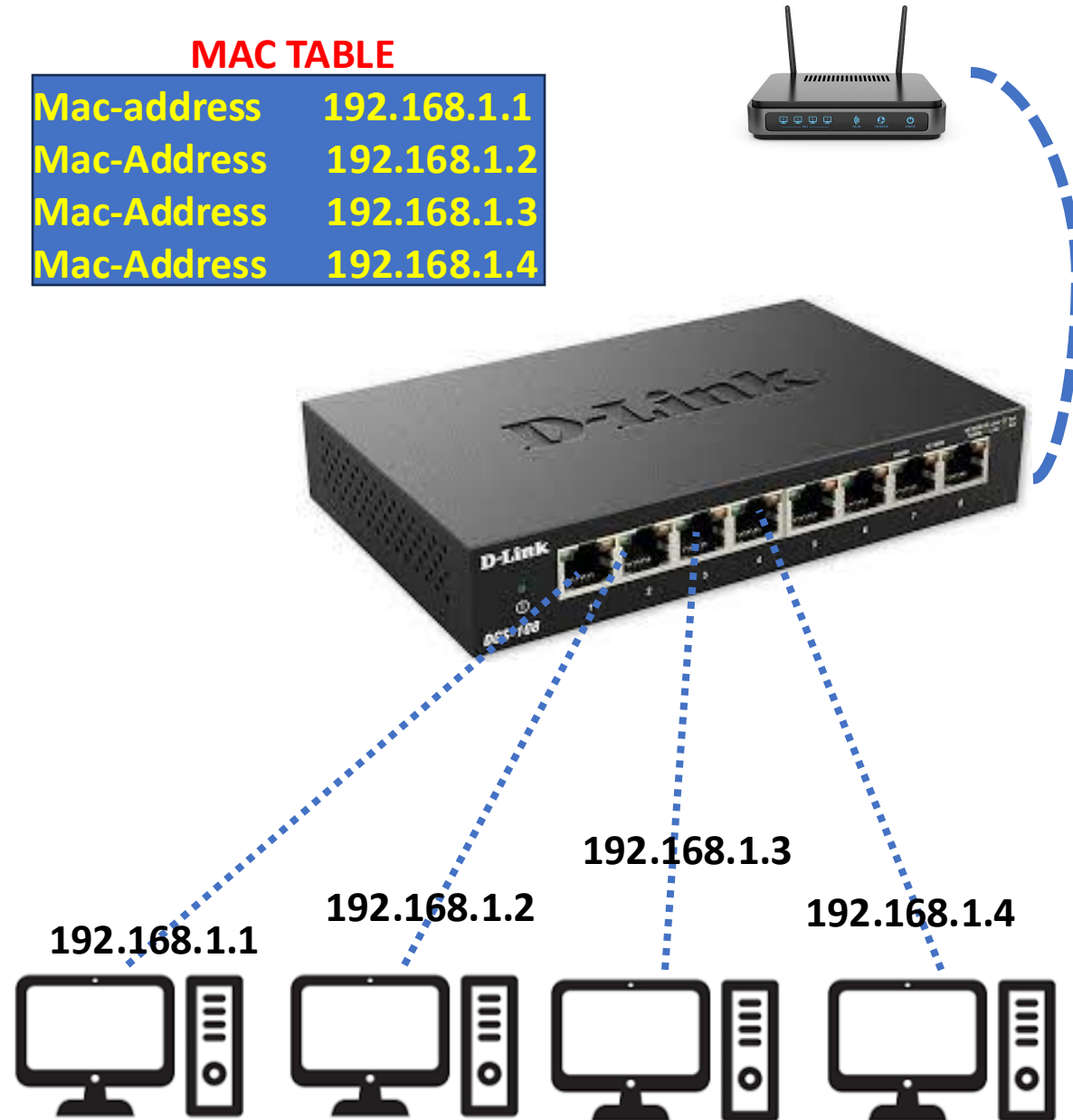
Switches make networks faster by avoiding unnecessary data traffic, unlike older hubs that send data everywhere.

Works on Which Layer?

OSI Layer 2 (Data Link: uses MAC addresses to direct data).

Real-World Example

In an office, a switch connects multiple computers to share files. In RHEL, list devices on the network with: `arp -a`.



Router

What is it?

A router is a device that directs data between different networks, like guiding traffic from your home network to the internet.

Interesting Use (Why it's cool)

Routers let you connect to websites worldwide by finding the fastest path for your data. Nowadays, routers provide an Access Point facility.

Works on Which Layer?

OSI Layer 3 (Network: uses IP addresses to route data).

Real-World Example

Your router creates a LAN for the office, team, or house with a single point of connection.



Access Point (AP)

What is it?

An access point creates a Wi-Fi network, letting wireless devices like phones connect to a wired network.

Interesting Use (Why it's cool)

APs let you use Wi-Fi in public places like cafes, turning a wired internet connection into a wireless one!

Works on Which Layer?

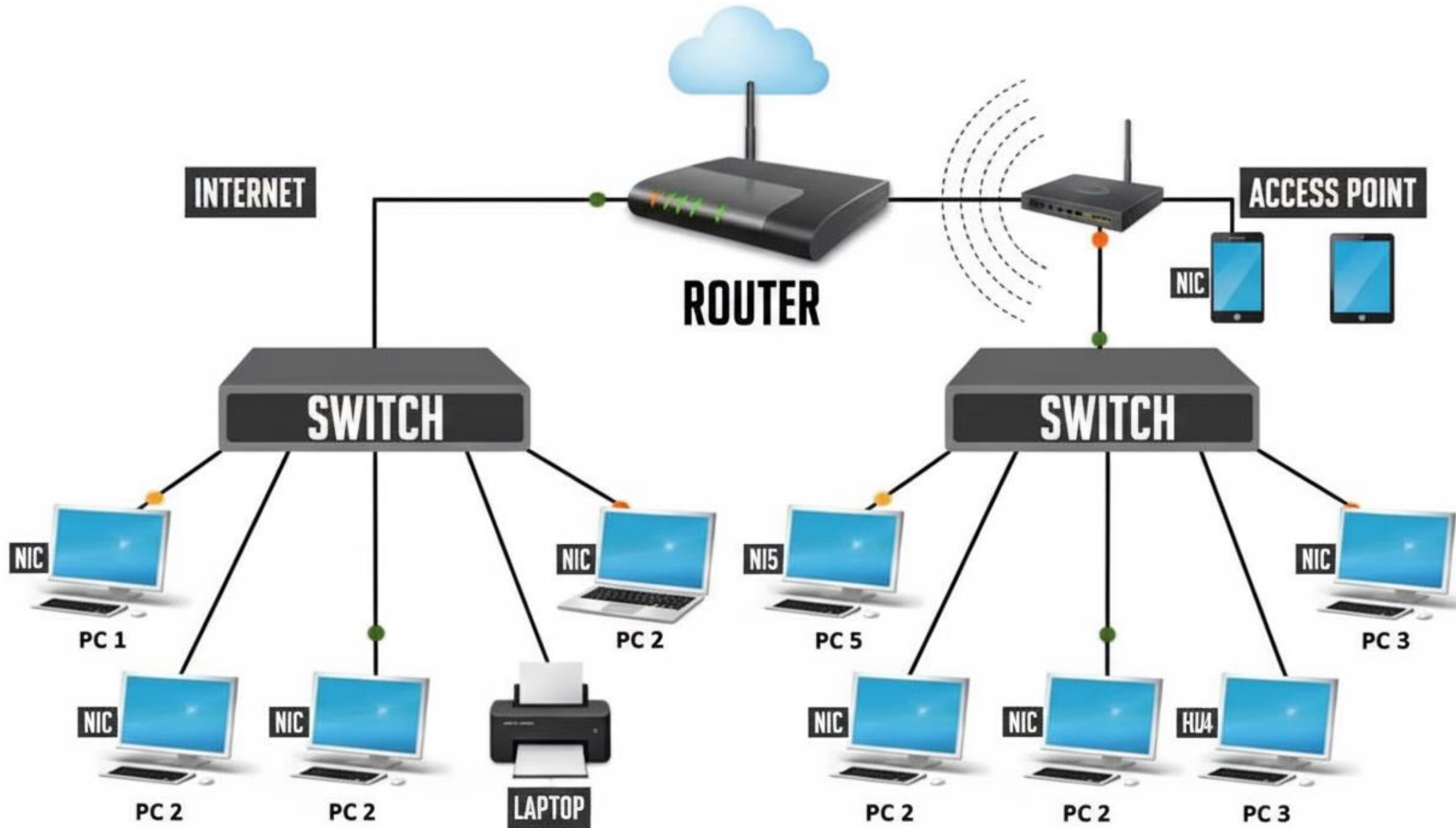
OSI Layer 2 (Data Link: manages wireless data transfer).

Real-World Example

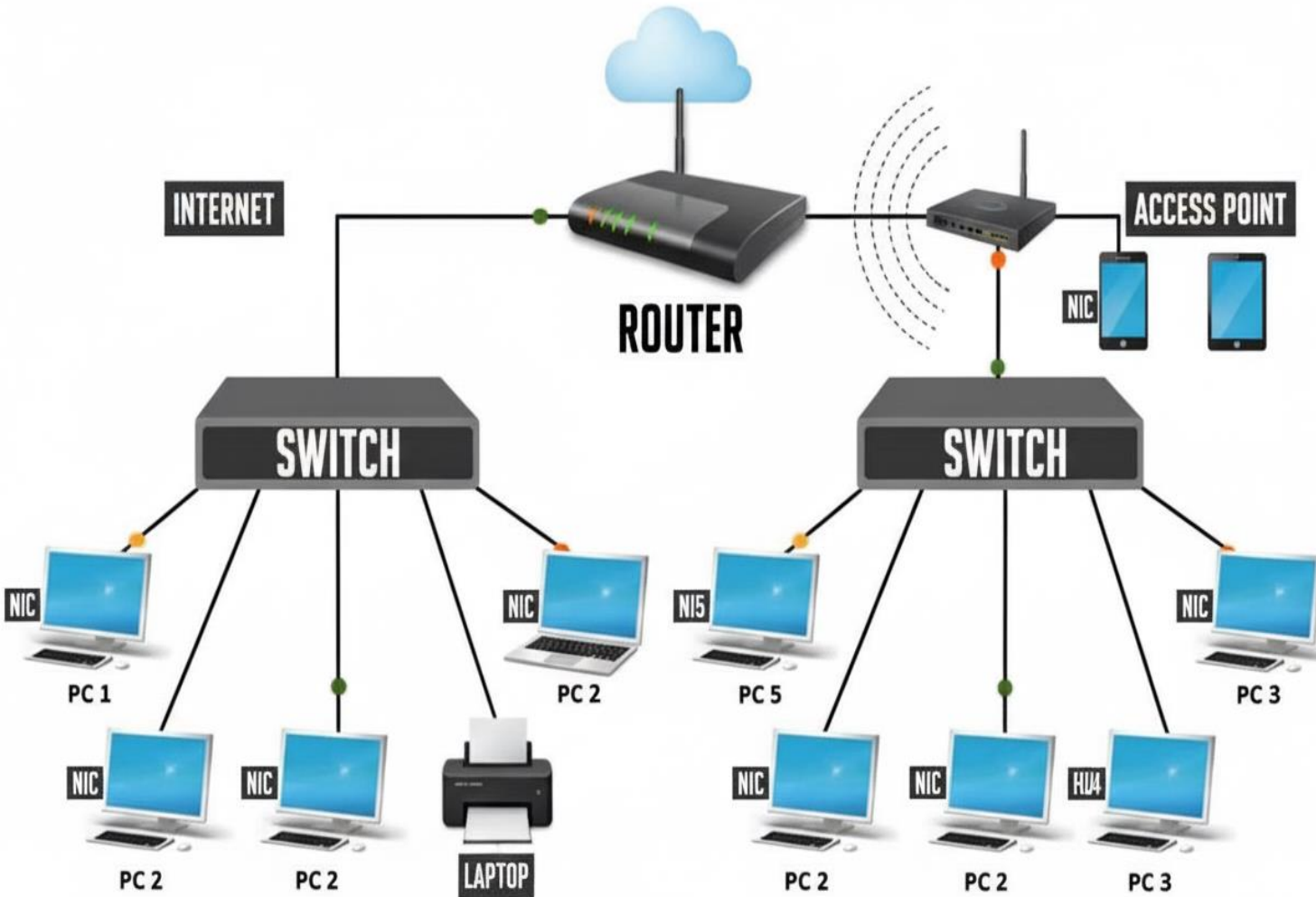
A coffee shop's Wi-Fi access point lets customers connect. In RHEL, join a Wi-Fi network with: `nmcli dev wifi connect "SSID" password "your_password"`.



Image: A complete diagram of networking in action.



What is LAN (Local Area Network) ?



- **Full form:** Local Area Network
- **Simple meaning:**
A **LAN** is a small network that connects computers and devices **inside one building or a small area** — like your **home, school, or office**.
- **Example:** When you connect your **laptop and phone to the same Wi-Fi**, they are part of one LAN.
- **Why we use LAN:**
 - To **share files**, printers, or internet easily.
 - To **play games** or send data between nearby computers.
- **Speed:** Very **fast**, because all devices are close to each other

LAN = Local Area Network



- End Devices in the LAN cannot communicate outside with their private IP.
- To access the internet, devices need a public IP, which is usually assigned to your router.

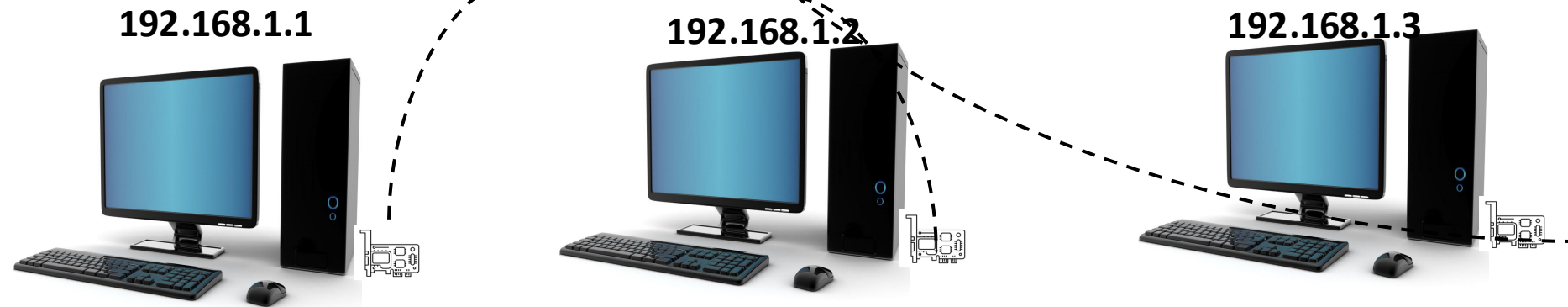


NAT

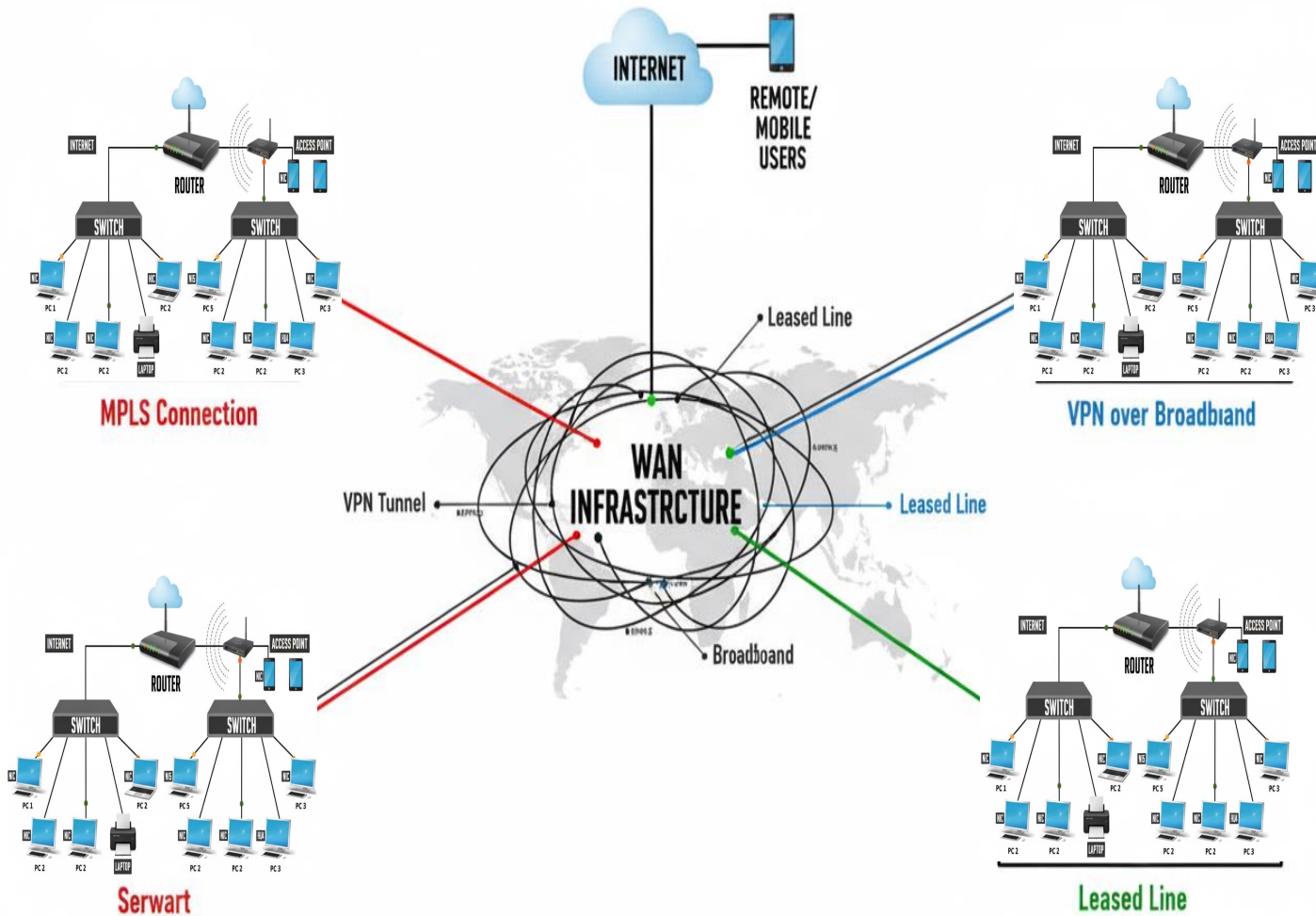
Your Router has a Public IP (Internet surfing IP)

- This Router works as a gateway device for your LAN-connected end-devices.
- The router has a Public IP address from the ISP (Airtel, Jio, Excitel).
- All connected devices are connected to a router, known as the LAN system

Localhost (LAN) = Same router connected devices known as LAN devices



What is WAN (Wide Area Network) ?



WAN = Many LAN Together

- **Full form:** Wide Area Network
- **Simple meaning:**
A WAN is a big network that connects many LANs together over a large distance — like between different cities or countries.
- **Example:** The Internet is the biggest WAN in the world.
A company that connects its Delhi office and Mumbai office using leased lines or internet is using a WAN.
- **Why we use WAN:**
 - To connect people far away.
 - To share information across locations.
- **Speed:** Slower than LAN, because it travels long distances.

Networking Addressing Related Concepts

- MAC address (Physical Address)
- IP address (IPv4 & IPV6 basics)
- Types of IP (public, private, loopback, etc.)
- Static vs dynamic IP
- Network address & broadcast address
- Gateway
- Ports & sockets
- Server-client model

MAC Address (Physical Address)

What is it? A unique identifier for a network device's NIC, like a device's "fingerprint."

Interesting Fact (Use) MAC address helps to ensure data reaches the correct device in a local network, like addressing a letter to a specific house.

Layer it Works On Data Link Layer (Layer 2) of the OSI model.

Real-World Example Your laptop's NIC uses its MAC address (e.g., 00:1A:2B:3C:4D:5E) to communicate with a router & Switches.

RHEL Command: `ipconfig /all`

```
C:\Users\Rakesh-PC>ipconfig /all_
```

```
Wireless LAN adapter Wi-Fi 2:
```

```
Connection-specific DNS Suffix . : bbrouter
Description . . . . . : TP-Link Wireless Nano USB Adapter
Physical Address. . . . . : 24-2F-D0-D9-C1-CB
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::e3b1:c6:353c:e5a0%24(Preferred)
IPv4 Address. . . . . : 192.168.1.36(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 October 2025 09:13:55 AM
Lease Expires . . . . . : 10 October 2025 09:13:56 AM
Default Gateway . . . . . : fe80::1%24
                             192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 354693072
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-19-99-DF-A0-8C-FD-FE-9F-68
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

IP Address (Logical Address)

What is it? A numerical address assigned to a device for network communication, like a phone number.

Interesting Fact (Use): Allows devices to find each other across networks, enabling internet access or file sharing.

Layer it Works On Network Layer (Layer 3) of the OSI model.

Real-World Example Your computer's IP (e.g., 192.168.1.100) lets it access websites via the internet.

RHEL Command Check IP: `ip addr show` (look for inet 192.168.1.100/24).

```
C:\Users\Rakesh-PC>ipconfig /all_
```

Wireless LAN adapter Wi-Fi 2:

```
Connection-specific DNS Suffix . : bbrouter
Description . . . . . : TP-Link Wireless Nano USB Adapter
Physical Address. . . . . : 24-2F-D0-D9-C1-CB
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e3b1:c6:353c:e5a0%24(Preferred)
IPv4 Address. . . . . : 192.168.1.36(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 October 2025 09:13:55 AM
Lease Expires . . . . . : 10 October 2025 09:13:56 AM
Default Gateway . . . . . : fe80::1%24
                             192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 354693072
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-19-99-DF-A0-8C-FD-FE-9F-68
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

IPv4 (Internet Protocol V4):

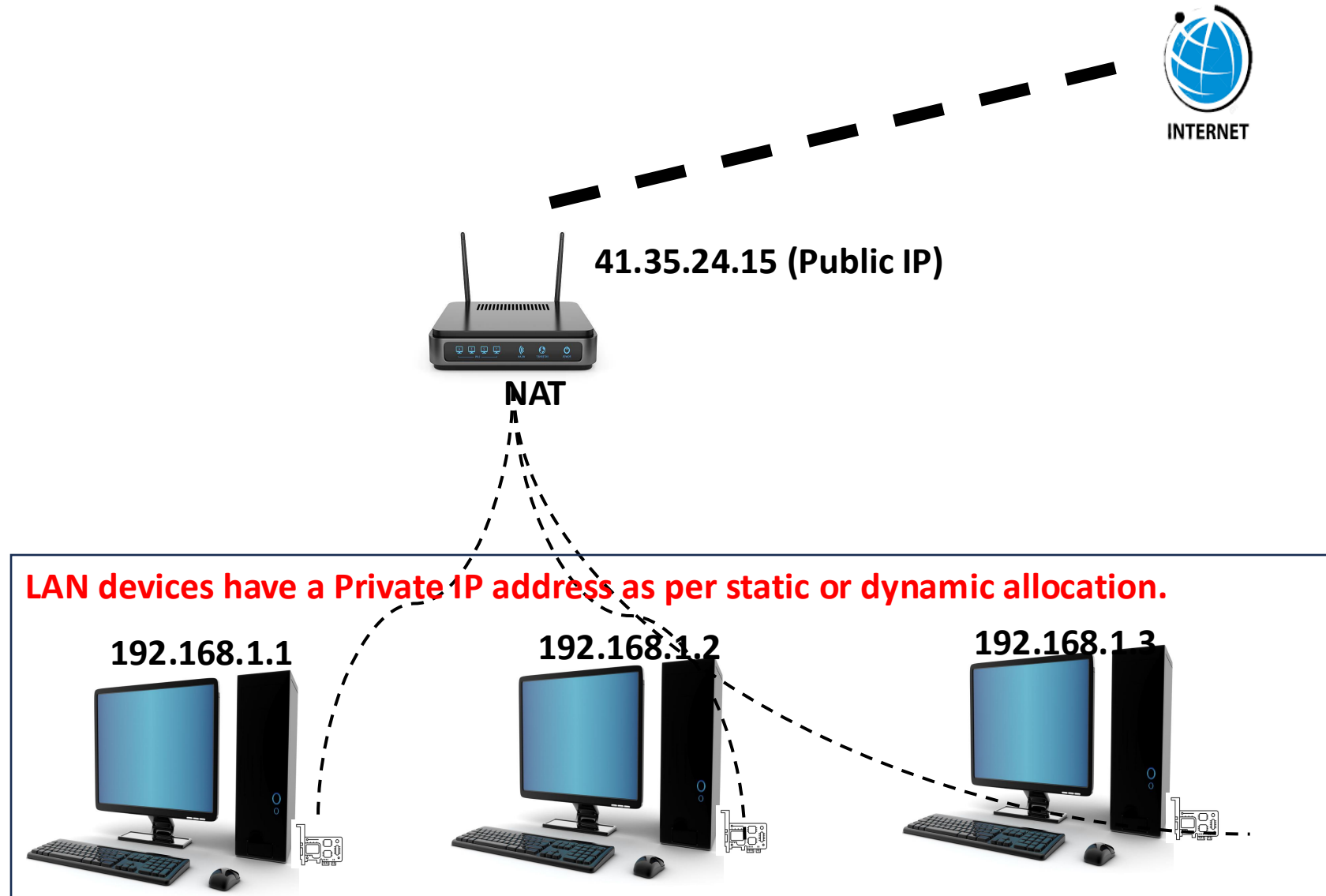
- 32-bit address (e.g., 192.168.1.1).
- Limited to ~4.3 billion addresses.
- Simple but running out of addresses, leading to IPv6 adoption.
- Total Addresses: Around 4.3 billion unique IPs (2^{32})
- Limitation: The number of available addresses is limited, which led to the creation of IPv6.

IPv6 (Internet Protocol V6):

- 128-bit address (e.g., 2001:0db8::1).
- Supports a massive number of addresses (~340 undecillion).
- Offers better security and auto-configuration features.
- Total Addresses: About 3.4×10^{38} (a huge number — enough for every device on Earth and more!)
- Advantages over IPv4:
 - Larger address space (no shortage).
 - Built-in security (IPSec)
 - Auto-configuration support
 - Better speed and efficiency

Types of IP Address (based on usage (where they're used):

- **Public IP:** Your Networking Device (Router/Gateway) IP, provided by your ISP (Airtel, Jio, Excitel, etc) to surf or travel over the internet. Without a public IP, your device can't travel on the internet.
- **Private IP:** The IP address assigned to devices within your local network (LAN), such as your computer, phone, or printer. Private IPs are not accessible directly from the Internet
- These are usually provided by your **router's DHCP server** for internal communication or directly by static request.



How to Confirm Which One Is a Public IP or Private IP

Remember, only these 3 ranges are private

The Only Private IP Range is –

10.0.0.0 – 10.255.255.255 (10/8)
172.16.0.0 – 172.31.255.255 (172.16/12)
192.168.0.0 – 192.168.255.255 (192.168/16)

These three blocks are the only private blocks in IPv4. All IP ranges *outside these 3 blocks* are public

All IP ranges *outside these 3 blocks* are public.

But a company **cannot** choose any random public IP block.

Why?

Because **public IPs are owned by authorities like APNIC, ARIN, RIPE**. So Authorities provide a public IP range as per our need.

You simply request to your **ISP / Internet Authority**:

- “We need 50 public IPs”
- “We need 20 public IPs”
- “We need 2 static public IPs”
- “We need 1 public IP for our gateway”

They check availability and assign you a block like:

203.123.40.0/27 → **32 IP's Pool**

or

103.55.20.0/28 → **16 IP's Pool**

or

43.224.18.0/29 → **8 IP's Pool**

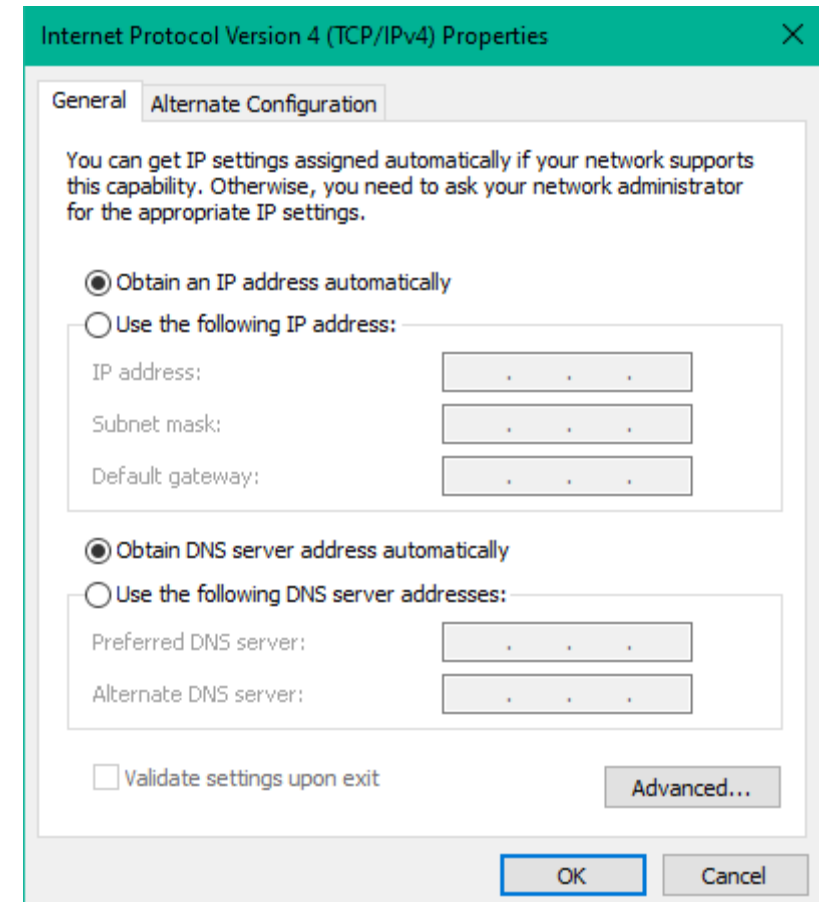
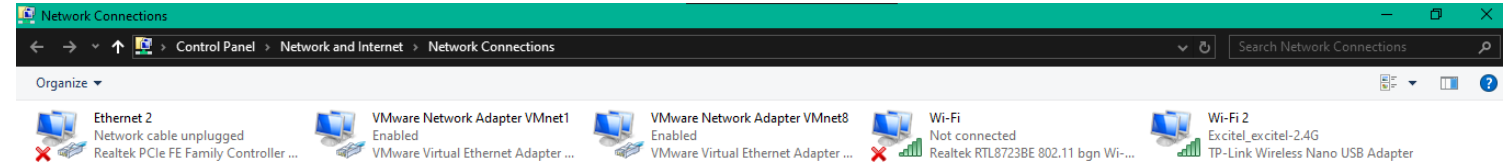
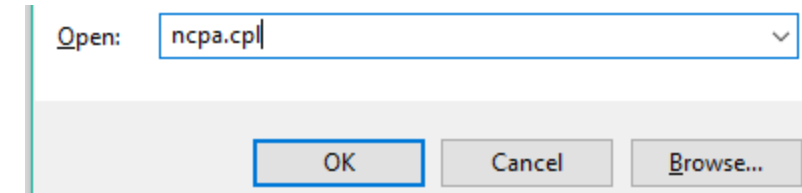
How an IP address is assigned to a device's Network Interface Card (NIC)

- **Static IP:**

A static IP address is manually assigned to a device and does not change automatically. It remains fixed until changed by the administrator — often used for servers, printers, and routers.

- **Dynamic IP:**

A dynamic IP address is automatically assigned by a DHCP (Dynamic Host Configuration Protocol) server. It can change every time the device connects to the network, commonly used for home and office computers.



6. Network Address

What it means:

The network address is the *starting or base address* of a network.

It represents the entire network — not any single device.

Example:

If your network is 192.168.1.0/24,
then all devices from 192.168.1.1 to 192.168.1.254
belong to this network.

Here, 192.168.1.0 is the **Network Address** — it's like the *name of the street* where all houses (devices) live.

Run command: ip addr show

7. Broadcast Address

What it means:

The broadcast address is a *special IP address* used to send data to **all devices** in the same network at once.

Example:

In the same network 192.168.1.0/24,
the broadcast address is 192.168.1.255.

If any device sends a message to 192.168.1.255,
every device in that network receives it —
just like a teacher announcing to the whole class.

Run command: ip addr show

8. Gateway

What is it? A device (usually a router) that connects a local network to external networks, like the internet.

Interesting Fact (Use) Acts like a “door” for your devices to access the outside world, such as Google’s servers.

Real-World Example Your home router (e.g., 192.168.1.1) is the gateway for your laptop to reach the internet.

RHEL Command Check gateway: `ip route` (e.g., default via 192.168.1.1); set: `nmcli con mod "eth0" ipv4.gateway "192.168.1.1"`.



8. Server-Client Model

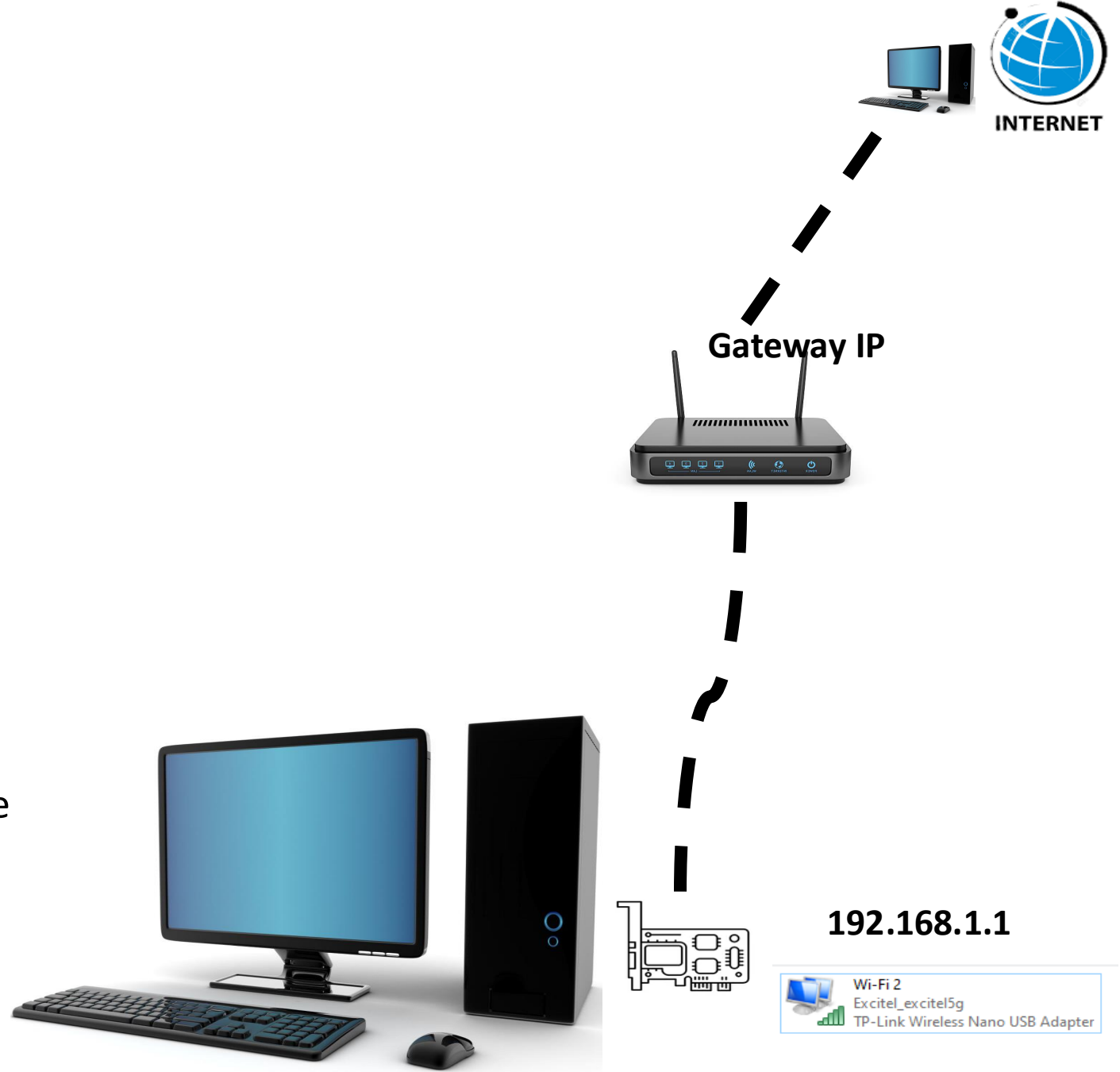
What is it? A network model where a **server** provides services (e.g., web pages) to **clients** (e.g., your browser).

Interesting Fact (Use) Powers most internet services, like streaming Netflix (server) on your phone (client).

Layer it Works On Application Layer (Layer 7), but relies on lower layers for transport.

Real-World Example A web server (e.g., Apache on RHEL) serves a website to your browser (client) when you visit a URL.

RHEL Command Check server status: `systemctl status httpd` (for Apache web server).



9. Ports and Sockets

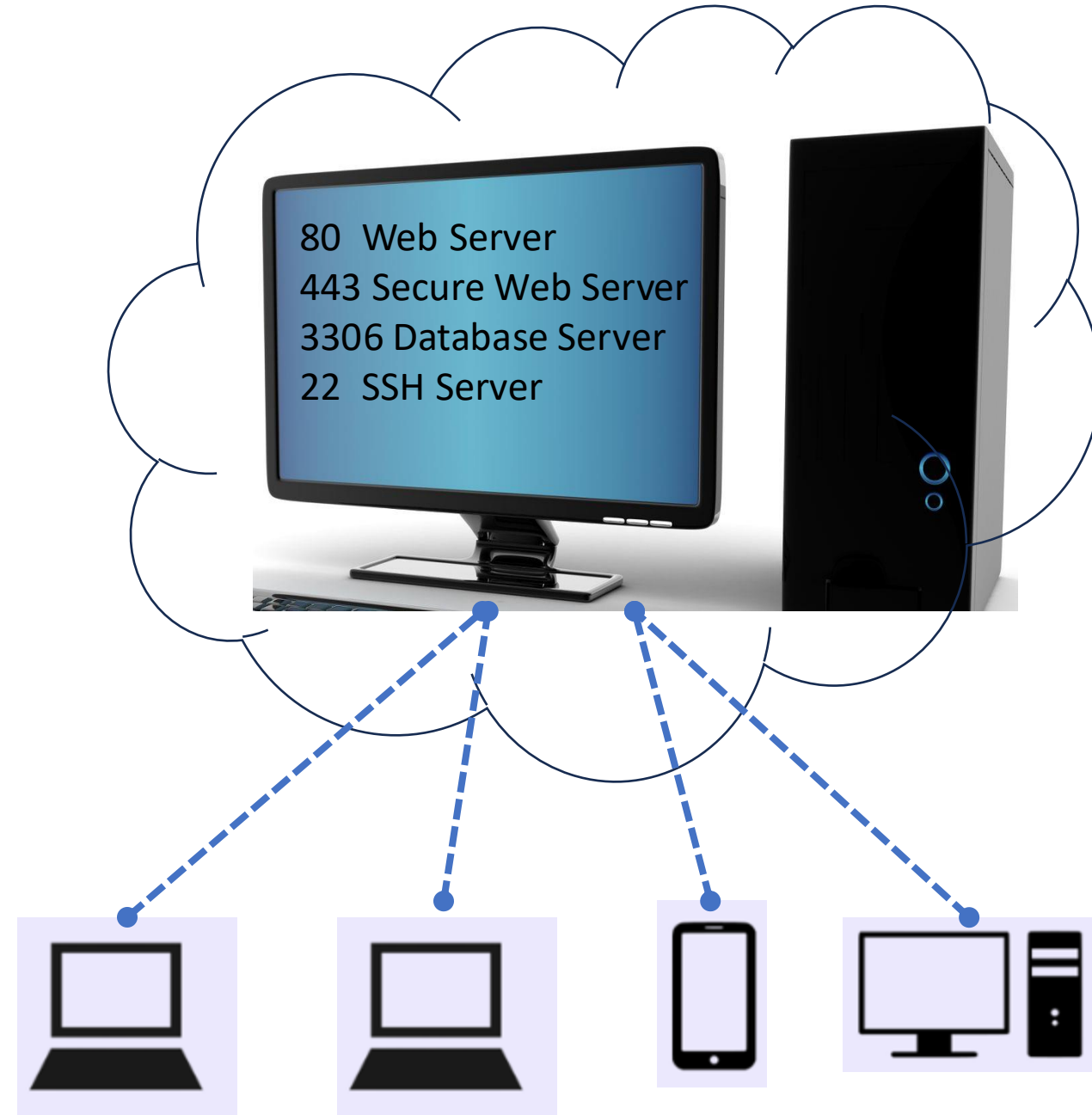
What is it?

Ports: Ports are Virtual “doors” in the server, where your service is running (numbers, e.g., 80 for HTTP) on a device for specific services.

Sockets: A combination of IP and port (e.g., 192.168.1.100:80) for communication.

Real-World Example A web server uses port 80; your browser connects to 192.168.1.100:80 to load a website.

RHEL Command Check open ports: `ss -tln` (lists ports like 80, 22); open port: `firewall-cmd --add-port=80/tcp --permanent`.



Ports are necessary to open on the server to provide service.

SERVER SERVICES



Socket = Server IP + Port

Suppose Server IP is 43.25.36.52

Service Name	Listen On Port/Protocol	Socket
http WebServer	80/tcp	43.25.36.52:80
https web server secure	443/tcp	43.25.36.52:443
Database Server	3306/tcp	43.25.36.52:443
DNS Server	53/udp	43.25.36.52:53
SSH Server	22/tcp	43.25.36.52:22

TCP (Transmission Control Protocol)

What is it? A protocol that makes sure data is sent and received correctly, like a reliable mail service.

Cool Fact (Use) It's like sending a package with tracking to ensure it arrives without errors, great for emails or websites.

Layer it Works On Transport Layer (Layer 4), handling data delivery.

Real-World Example Loading a webpage (e.g., Google on port 80) uses TCP to get all page parts without missing anything.





UDP (User Datagram Protocol)

What is it? A fast protocol that sends data quickly without checking if it arrives, like a quick text message.

Cool Fact (Use) Perfect for live video calls or gaming where speed is more important than perfect data.

Layer it Works On Transport Layer (Layer 4), focusing on fast delivery.

Real-World Example Watching a live YouTube stream uses UDP (e.g., port 123 for time sync) to avoid delays.

Important Networking Commands

Check MAC Address: `ip link show`.

Example output: `link/ether 00:1A:2B:3C:4D:5E`.

Check IP Address: `ip addr show`.

Example output: `inet 192.168.1.100/24`.

Example: `tcp LISTEN 0 128 *:80 *:*`.

Check Gateway: `ip route show | grep default`.

Example output: `default via 192.168.1.1`.

Check Public IP:

`curl ifconfig.me`

Example output: `203.0.113.1`.

Check Private IP:

`ip addr show | grep inet`

Example output: `inet 192.168.1.100/24`.

Check IPv6:

`ip -6 addr`

Example output: `inet6 2001:0db8::1/64`

Check DHCP IP Assignment:

`nmcli device show | grep IP4.ADDRESS`

Check Gateway:

Example: `ip route`

Example: `route -n`

ip:

Checks interfaces, routing, and addresses.

Example: `ip addr show` (shows IP addresses).

ifconfig:

Older tool for interface configuration.

Example: `ifconfig eth0`.

ping:

Tests connectivity.

Example: ping google.com.

/etc/hosts:

Maps hostnames to IP addresses.

Example content:

192.168.1.1 myserver

/etc/resolv.conf:

Configures DNS servers.

Example content:

nameserver 8.8.8.8

nameserver 8.8.4.4

nslookup:

Queries DNS.

Example: nslookup google.com.

traceroute/tracepath:

Traces the path of data packets.

Example: traceroute google.com.

netstat:

Older tool to check network connections.

Example: netstat -tuln.

ss:

Modern tool for network connections.

Example: ss -tuln.

wget:

Downloads files from the web.

Example: wget <http://example.com/file.txt>.

curl:

Fetches data from the web.

Example: curl http://example.com.

Subnetting

(To distribute the IPv4 IP range in the network, we use subnetting)

And subnetting is one of two types. Class-based and Classless Subnetting

What is Subnetting?

- **Subnetting** is the process of **dividing a large network into smaller, manageable networks** called **subnets**.
- Think of it like **splitting a big village into smaller streets**, so it's easier to find houses and manage traffic.
- Helps **efficiently use IP addresses** and **improves network performance and security**
- Modern subnetting is done using **CIDR** because classful masks are no longer used

Why do we need Subnetting?

- **Better organization:** Large networks can be broken into smaller parts.
- **Efficient IP usage:** Avoid wasting IP addresses.
- **Improved security:** Traffic can be isolated between subnets.
- **Better performance:** Less congestion because smaller networks reduce broadcast traffic.

Classful VS Classless Subnetting

1. **Classful Subnetting** (Old Method- Depreciated, not used in modern networking.)

What it means:

Subnetting is done **inside fixed IP classes** (Class A, B, C).

You are **not allowed** to change the default class boundaries.

Default classes:

- **Class A** → 0.0.0.0 – 127.255.255.255 ---→ /8
- **Class B** → 128.0.0.0 – 191.255.255.255 ---→ /16
- **Class C** → 192.0.0.0 – 223.255.255.255 ---→ /24

So if you had: **Class C network** → 192.168.1.0 & **Default mask** = /24

In classful subnetting, your network **must** start with /24.

But you **cannot** change the network to something like /22 or /19 — because those do not align with the fixed class boundary.

Classful VS Classless Subnetting

What are the Problems with classful subnetting

- Wasted IPs
- Fixed blocks too big or too small
- No route aggregation (routing tables became huge)

This method is **not used today**.

So, now, for **classless** subnetting, in modern times, **CIDR** is used.

Classful VS Classless Subnetting

2. Classless Subnetting (Modern Method — CIDR)

What it means:

No restriction of Class A/B/C boundaries.

You can use **any prefix** for any network.

Examples:

- 10.0.0.0/8
- 10.0.0.0/12
- 172.16.0.0/20
- 192.168.1.0/23
- 100.64.0.0/22
- 203.0.113.0/27

Everything is flexible.

This is enabled by **CIDR (Classless Inter-Domain Routing)**.

Classful VS Classless Subnetting

Advantages of Classless Subnetting

- Super efficient IP usage
- Better subnetting
- Route summarization
- Modern routers and ISPs only use this

This is the **current standard** everywhere.

How to device Classless Subnetting

Simply understand this CIDR table

Subnet Prefix	Host Range
/19	8192
/20	4096
/21	2048
/22	1024
/23	512
Any-Random-IP/24	256
/25	128
/26	64
/27	32
/28	16
/29	8
/30	4
/31	2
/32	1

Now any IP range has:

Network Address: Smallest IP in your Range

Broadcast Address: Highest IP in your Range

Subnet Prefix	Host Range
/19	8192
/20	4096
/21	2048
/22	1024
/23	512
Any-Random-IP/24	256
/25	128
/26	64
/27	32
/28	16
/29	8
/30	4
/31	2
/32	1

How to Confirm Which One Is a Public IP or Private IP

Remember, only these 3 ranges are private

The Only Private IP Range is –

10.0.0.0 – 10.255.255.255 (10/8)
172.16.0.0 – 172.31.255.255 (172.16/12)
192.168.0.0 – 192.168.255.255 (192.168/16)

These three blocks are the only private blocks in IPv4. All IP ranges *outside these 3 blocks* are public

All IP ranges *outside these 3 blocks* are public.

But a company **cannot** choose any random public IP block.

Why?

Because **public IPs are owned by authorities like APNIC, ARIN, RIPE**. So Authorities provide a public IP range as per our need.

You simply request to your **ISP / Internet Authority**:

- “We need 50 public IPs”
- “We need 20 public IPs”
- “We need 2 static public IPs”
- “We need 1 public IP for our gateway”

They check availability and assign you a block like:

203.123.40.0/27 → **32 IP's Pool**

or

103.55.20.0/28 → **16 IP's Pool**

or

43.224.18.0/29 → **8 IP's Pool**

Good Day!